

CS23532-COMPUTER NETWORKS-LAB MANUAL

Practical -4

Aim

To set up and configure a Local Area Network (LAN) using a switch and Ethernet cables.

Theory

A **LAN (Local Area Network)** connects computers within a limited area, such as a laboratory, office, or home. A switch connects the computers and enables them to communicate and share resources.

Requirements

- Computers (3–4)
- Network switch
- Ethernet cables

Procedure

1. Connect 3–4 computers to a switch using Ethernet cables.
2. Assign IP addresses and subnet masks to each computer.
3. Verify the IP configuration using the ipconfig command.
4. Check connectivity between computers using the ping command.
5. Share a folder on one computer and access it from another computer on the same LAN.

IP Address Configuration

Device IP Address Subnet Mask

PC1	10.1.1.1	255.0.0.0
PC2	10.1.1.2	255.0.0.0
PC3	10.1.1.3	255.0.0.0
PC4	10.1.1.4	255.0.0.0
Switch	10.1.1.5	255.0.0.0

Verification

1. Check IP configuration

ipconfig

2. Test connectivity from PC1 to PC2

ping 10.1.1.2

Expected output: Replies are received from the destination computer if connectivity is successful.

The image shows a screenshot of the 'Internet Protocol Version 4 (TCP/IPv4) Properties' dialog box, specifically the 'General' tab. The dialog box has a title bar with the text 'Internet Protocol Version 4 (TCP/IPv4) Properties' and a close button (X) in the top right corner. The 'General' tab is selected, and it contains the following elements:

- A text box with the message: 'You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.'
- Two radio buttons for IP address assignment:
 - ☐ Obtain an IP address automatically
 - ☒ Use the following IP address:
- Fields for manual IP configuration (enclosed in a box):
 - IP address: 10 . 1 . 1 . 1
 - Subnet mask: 255 . 255 . 255 . 0
 - Default gateway: . . .
- Two radio buttons for DNS server address assignment:
 - ☐ Obtain DNS server address automatically
 - ☒ Use the following DNS server addresses:
- Fields for manual DNS configuration (enclosed in a box):
 - Preferred DNS server: . . .
 - Alternate DNS server: . . .
- A checkbox labeled 'Validate settings upon exit' which is currently unchecked.
- An 'Advanced...' button located to the right of the 'Validate settings upon exit' checkbox.
- At the bottom of the dialog box are 'OK' and 'Cancel' buttons.

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 10 . 1 . 1 . 2

Subnet mask: 255 . 255 . 255 . 0

Default gateway: . . .

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel

Result

A LAN was configured using a switch and Ethernet cables. IP addresses were assigned, connectivity was tested using ping, and file sharing was performed between computers.

